

Contents

The deposition-only stellar mass model: setup, options, scoring, results	1
1. What the model is trying to do	1
2. The model, in full	1
3. The options we tried, explained	3
4. How models are scored	5
5. The main result	6
6. The progenitor-selection confound, and why it mattered	9
7. Two defects in the scoring, and what they changed	11
8. What we have learned	12
9. Where the model stands, honestly	14
Appendix: where things live	14

The deposition-only stellar mass model: setup, options, scoring, results

Status as of 2026-08-23, after exp54 Stages 0–3.4. Supersedes the summaries in `experiments/exp54_unpinned_amplitude` which this document consolidates and explains.

This note is meant to be readable by someone who has not been in the room. Every symbol, every code name (E2, S3, `gomper_tz_log`), and every metric is defined where it first appears. Where a name is a piece of internal shorthand rather than a standard term, that is said explicitly.

1. What the model is trying to do

The question. Given only the *dark matter halo*’s history — how its mass grew over cosmic time, how big it was, how concentrated — can we predict how much stellar mass a massive galaxy has, and how that stellar mass is distributed with radius, at five different cosmic epochs?

The data. 2397 galaxies from the TNG300 cosmological simulation, selected at redshift $z = 0.4$ with stellar masses $\log_{10} M_*$ between 10.66 and 12.36. For each galaxy we have:

- its **mass accretion history** (MAH): the halo mass $M_h(t)$ at 73 snapshots, from the smooth DiffMAH fit;
- its **halo structure** at 16 snapshots: R_{200c} , the NFW concentration c_{200c} ;
- its **curve of growth** (CoG) at five epochs — see below.

Curve of growth (CoG), defined. $M_*(<R)$, the total stellar mass inside a sphere of radius R , measured on a fixed 24-point radial grid from 2 kpc to 148.22 kpc. It is cumulative, so it only ever rises with R . This is the thing the model must reproduce.

The five epochs. $z = 0.4, 0.7, 1.0, 1.5, 2.0$ — snapshots 72, 59, 50, 40, 33. At the four higher redshifts we follow each galaxy’s **main progenitor**, the most massive branch of its merger tree.

“Deposition-only”, defined. The model builds a galaxy by *depositing* stellar mass at each step of the halo’s growth and never moving it again. Each deposit is a small blob of stars laid down with a fixed radial profile, and the galaxy at any later time is the sum of all deposits laid down up to then. There is no subsequent transport, no re-arrangement, no dynamical evolution. That is a strong assumption, and it is the point: it is the simplest thing that could work, and Section 8 reports where it breaks.

2. The model, in full

2.1 The three equations

For one galaxy, let t_j index the steps of its halo’s mass accretion history (one per simulation snapshot, ~ 72 steps). At step j the halo gains mass $dM_{h,j}$ and has properties $H_j = (\text{mass, redshift, radius, concentration,}$

...).

(1) **The budget** — how much stellar mass this step deposits:

$$\Delta M_{*,j} = \varepsilon_{\text{surv}}(H_j) \Delta M_{h,j}$$

(2) **The boundary** — the outer edge of the deposit:

$$R_{\text{trunc},j} = C R_{200c}(t_j), \quad C = 3 \text{ throughout}$$

(3) **The sum** — the galaxy’s curve of growth at observation epoch t_k :

$$M_*(< R, t_k) = \sum_{j: t_j \leq t_k} \Delta M_{*,j} F_j(< R)$$

$F_j(< R)$ is a **unit-mass cumulative profile**: a function rising from 0 to 1 that says what fraction of deposit j ’s mass lies inside radius R . It is truncated at $R_{\text{trunc},j}$ and renormalised so that $F_j(R_{\text{trunc},j}) = 1$ exactly.

That is the whole model. Three lines.

2.2 What $\varepsilon_{\text{surv}}$ is, and what it is not

$\varepsilon_{\text{surv}}$ is internal shorthand for the **effective surviving-central mass fraction**: the fraction of the halo mass gained in a step that ends up as stellar mass still bound to the central galaxy at the epoch we observe it.

It is not a star-formation efficiency, and it must not be quoted as one. It silently absorbs at least five distinct physical processes:

1. conversion of accreted baryons into stars,
2. stellar evolutionary mass loss (stars die and return gas),
3. tidal stripping of the stars that do form,
4. whether the stars end up in the central galaxy or in a satellite,
5. delivery of stars formed elsewhere and later accreted (*ex-situ* mass).

Because it is a single number standing in for all five, **no individual physical interpretation of its value is legitimate**. What *is* legitimate is its dependence on halo mass and redshift, since that dependence is measured.

2.3 Why there is a truncation, and why it matters more than it looks

Without a boundary, “the total mass of a deposit” is a mass at infinity, which for some profile shapes is a badly behaved quantity. Concretely: the fitted Moffat profile falls off as $R^{-0.76}$, so **a deposit with a 50 kpc half-mass radius needs to extend to 9.6 Mpc to enclose 99% of its mass**. That is not a galaxy. It caused 18% of the model’s stellar mass to sit beyond 500 kpc in an earlier version.

Truncating at $C * R_{200c}(t_j)$ fixes this: the deposit’s “total” is the mass inside a *halo-defined* radius, which is finite, physical, and epoch-local.

$C = 3$ is a **working default, not a physical law**. It was chosen because $R_{99} \leq 3 R_{200c}$ holds for the fitted profiles, and it was scanned rather than assumed.

The subtlety that had to be handled carefully. Cutting a profile off moves its half-mass radius *inward* — half the remaining mass is now inside a smaller radius than half the original mass was. So if you truncate a profile but keep labelling it by its *untruncated* half-mass radius, you are no longer measuring the same quantity, and comparing two differently-shaped families becomes meaningless. Measured mislabelling: **+10% to +35%**.

The model therefore parameterises by the **true post-truncation half-mass radius**. Writing F_0 for the untruncated unit profile in units of its own half-mass radius (so $F_0(1) = 0.5$) and $u = R_{\text{trunc}} / R_{50_{\text{untruncated}}}$:

$$F_{\text{trunc}}(x) = \frac{F_0(x)}{F_0(u)}, \quad x_h(u) = F_0^{-1}\left(\frac{1}{2}F_0(u)\right), \quad h(u) = \frac{x_h(u)}{u} = \frac{R_{50}^{\text{true}}}{R_{\text{trunc}}}$$

h decreases monotonically, so it can be inverted: given the size law's requested R_{50_true} , the code solves $h(u) = R_{50_true} / R_{\text{trunc}}$ for u and recovers the family's own internal scale. A numerical self-check asserts $F(R_{50}) = 0.5$ and $F(R_{\text{trunc}}) = 1$ for every family to better than $5e-3$ and $1e-12$ respectively.

3. The options we tried, explained

The model has three independent axes. Each option is *nested* inside the richer one, so “does this extra term earn its place?” is a well-posed question.

Below, $m = \log_{10} M_h(t_j) - 13.5$ (halo mass at the deposit's own formation time, measured relative to a pivot of $10^{13.5}$ solar masses) and $x = \ln(1 + z_j)$ (a redshift variable that is 0 today and grows with lookback time).

3.1 The efficiency axis — how much mass a step deposits

These control $\log_{10} \text{eps}_{\text{surv}}$. **The E-numbers are internal labels, not standard nomenclature.**

label	what it means in words	formula	free params
E1	Efficiency is a power law in halo mass and in $(1+z)$ — the simplest thing that could work.	$a_0 + a_M m + a_z x$	3
E2	E1 plus a mass×redshift cross-term: how efficiency depends on halo mass is itself allowed to change with redshift.	$a_0 + a_M m + a_z x + a_{Mz} m x$	4
E3	An <i>add-on</i> to E1 or E2: efficiency also depends on how concentrated the halo is, relative to typical for its mass and redshift.	$\dots + a_c d \log c$	+1
E4	The redshift dependence is a free spline — a flexible wiggly curve in $\ln(1+z)$ with 4 knots, instead of a straight line. Built to test whether the efficiency peaks at some redshift.	$a_0 + a_M m + \text{spline}(x)$	6

label	what it means in words	formula	free params
E5	Efficiency is a double power law in halo mass : rising as Mh^β below a characteristic mass $M1$, falling as $Mh^{-\gamma}$ above it, with a separate redshift power law. This is the shape abundance-matching studies find for the stellar-to-halo-mass relation.	$\log_{10} \epsilon_0 + \log_{10}(2) - \log_{10}(r^{-\beta} + r^\gamma) + a_z x$, where $r = Mh/M1$	5

dlogc, defined (used by E3 and S3): the **concentration excess**, $\log_{10}[c_{200c_measured} / c_{200c_Diemer19}(Mh, z)]$. It measures how much *more* concentrated a halo is than is typical for its mass and redshift. The reference $c_{200c_Diemer19}$ is a published deterministic formula, and subtracting it is what makes the variable informative: the raw concentration is almost entirely predictable from mass and redshift, so all of its independent signal lives in this residual.

3.2 The profile axis — what shape a single deposit has

Six functional forms for F_0 , the unit-mass cumulative profile of one deposit, all written so that R_{50} is genuinely the half-mass radius:

label	form	shape parameter
sersic	$\text{gamma} \text{inc}(2n, x^{(1/n)})$ — the standard galaxy profile	n (Sérsic index)
moffat	$1 - (1 + x^2)^{-(1-\text{gam})}$ — a power-law-tailed profile	gam
gompertz_log	$\exp(-\ln 2 * (R/R_{50})^{-c})$ — a sigmoid in log radius	c
loglogistic	$X/(1+X)$ with $X = (R/R_{50})^k$	k
richards	a generalisation containing both of the above	k, ν
expo, gauss	fixed-shape special cases	none

The factorial in Section 4 used the first three.

3.3 The size axis — how big a deposit is

These control R_{50} , the deposit's half-mass radius. f_0 and b are free.

label	what it means in words	formula
S2	The deposit's size is a fixed fraction of the halo's virial radius <i>at the time the deposit is laid down</i> , with a free power-law redshift tilt on top.	$R_{50} = f_0 * R_{200c}(t_j) * (1+z_j)^b$

label	what it means in words	formula
S2a	The same, but scaled to the halo's NFW scale radius R_{200c}/c_{200c} instead of R_{200c} — i.e. the deposit tracks the halo's inner structure rather than its outer edge.	$R_{50} = f_0 * [R_{200c}(t_j)/c_{eff}] * (1+z_j)^b$
S3	An <i>add-on</i> to either: the size normalisation is also allowed to depend on halo mass, concentration excess, and how early the halo formed.	$\log_{10} f = \dots + s_M * m + s_c * d \log c + s_f * (f_{form} - 0.5)$

f_form, defined: $t_{half}(t_j)/t_j$. In words — *of the mass this halo has right now, how long ago did it acquire half of it, as a fraction of its current age?* Near 0 means the halo assembled very early. Unlike other formation-time measures it never divides by the halo's *final* mass, so it can be computed at any epoch without knowing the future. That property is what makes it legal as a model input.

A naming trap worth flagging. The code reads `scale = R200c if size == "S2" else R200c/c_eff`. So **S2a takes the else branch and does use concentration** — a fact that was missed once and cost a wrong impact assessment (Section 7).

4. How models are scored

4.0 Which epochs are fitted

All five. The five-epoch fits (`stage33*.py`) minimise a single loss summed over $z = 0.4, 0.7, 1.0, 1.5$ and 2.0 simultaneously — `epochs=(0,1,2,3,4)`. There is no extrapolation in redshift: $z = 2$ is fitted like every other epoch.

The one place a reduced epoch set is used is the 45-cell **model-selection factorial** (`stage32*.py`), which fits only `epochs=(0,4)` — $z = 0.4$ and $z = 2.0$, the two **endpoints** — because it runs 45 models and needs to be affordable. There the interpolated epochs are $z = 0.7, 1.0$ and 1.5 , not $z = 2$. So at no stage is $z = 2$ an extrapolation; in the factorial it is one of only two epochs that constrain the fit at all.

A test worth running that has not been run. Fit only $z \leq 1.5$ and hold $z = 2.0$ out entirely, then score it. That converts $z = 2$ from a constraint into a genuine **extrapolation in redshift**, which is a much stronger claim about the model than fitting it: a physical law calibrated over $0.4 \leq z \leq 1.5$ that also predicts $z = 2$ has demonstrated something a fitted curve has not. It is cheap — one extra fit per model — and it is the natural way to advertise the model's capability rather than its flexibility. Recorded in `doc/todo.md`.

4.1 The split: amplitude and shape

Every predicted curve of growth is split losslessly into two independent parts:

$$A = \log_{10} M_*(< 100 \text{ kpc}) \quad (\text{amplitude: how much mass})$$

$$F(< R) = \frac{M_*(< R)}{M_*(< 100 \text{ kpc})} \quad (\text{shape: how it is distributed})$$

They are scored separately because they fail independently: a model can have the right total and the wrong distribution, or vice versa, and a single number cannot tell those apart.

4.2 The two scores

$$\text{score}_A = \sqrt{\left\langle \left(\frac{A_{\text{model}} - A_{\text{truth}}}{\sigma_A(z)} \right)^2 \right\rangle_{\text{galaxies, epochs}}} \quad \text{score}_F = \frac{\mathcal{L}(F_{\text{model}}, F_{\text{truth}}, R)}{L_F^{\text{ref}}}$$

Both are **normalised by a measured benchmark**, so that **1.0** means “as good as what we already know how to do”:

- **sigma_A(z)** is the scatter of the best *purely statistical* halo-only regression, refitted separately at each epoch — the best a regression can do using the same halo information. Its value is **0.1044 / 0.1120 / 0.1204 / 0.1279 / 0.1611 dex** at $z = 0.4 / 0.7 / 1.0 / 1.5 / 2.0$. So $\text{score}_A = 1.0$ means the physical model matches a regression; below 1 it beats one; above 1 it loses to one.
- **L_F_ref = 0.158196** is the previous-generation model’s shape loss.

The Objective used for shape is: fractional residuals ($\text{model} - \text{truth}$)/truth on the cumulative profile, root-mean-square over the 24 radii with equal weight, arithmetic mean over epochs, arithmetic mean over galaxies.

The fitted loss is $L = \text{score}_A^2 + \text{score}_F^2$, plus a penalty for any galaxy the model cannot represent at all (priced, never dropped — dropping failures makes a failing model’s loss silently improve).

4.3 The identifiability report — mandatory, and it has teeth

A parameter can be useless with a sharp optimum, or genuinely useful and completely undetermined, and the loss distinguishes neither. Every fit therefore ships:

- a scaled residual-Jacobian SVD, reporting **ident = smallest singular value / largest**. Small ident means some direction in parameter space barely changes the prediction — i.e. those parameters are not determined by the data.
- a profile scan per parameter, re-optimising the others;
- bootstrap stability over galaxies;
- profiles of *derived* quantities, which are often identified where their constituents are not.

A parameter that fails may stay in the model; its VALUE is never quoted.

4.4 The judge — and why we never rank by loss

Models are ranked by **mean rank across seven criteria**, not by the loss:

score_A at $z=0.4$ and $z=2.0$, score_F at $z=0.4$ and $z=2.0$, **ident**, an over-reach diagnostic, and the **halo-mass tilt**.

Halo-mass tilt, defined. For a fixed radius and epoch, take each galaxy’s residual $y = \log_{10} [M_{\star, \text{model}}(<R)] - \log_{10} [M_{\star, \text{truth}}(<R)]$, and fit a straight line of y against $\log_{10}(\text{halo mass})$ across the population. The tilt is that slope, in **dex of stellar-mass error per dex of halo mass**. Zero means the model is equally right for light and heavy haloes; non-zero means it has a systematic mass-dependent bias that the population-averaged loss hides.

This is not a stylistic preference. Ranking the 45-cell factorial by loss picks cells the judge places 2nd, 6th and 10th — and those loss-optimal cells have **ident** around $8e-04$ against the winner’s **7.8e-03**, an order of magnitude worse conditioned. **The loss rewards exactly the degenerate models.**

5. The main result

5.1 The adopted model

$$\log_{10} \varepsilon_{\text{surv}} = a_0 + a_M m + a_z x + a_{Mz} m x \quad (\text{E2})$$

$$R_{50} = f_0 R_{200c}(t_j) (1 + z_j)^b \quad (\text{S2})$$

$$F_0(R) = \exp[-\ln 2 (R/R_{50})^{-c}] \quad (\text{gompertz_log})$$

truncated at $3R_{200c}(t_j)$ and renormalised

Seven global parameters. No per-object freedom. No pinning to the truth. Fitted values on the full sample: $a_0 = -2.549$, $a_M = -0.452$, $a_z = 0.488$, $a_{Mz} = 0.233$, $\log f_0 = -0.952$, $b = -0.735$, $c = 0.919$.

It is the judge's rank-1 model out of 45, and it stayed rank 1 after two independent corrections to the scoring (Section 7).

5.2 What it gets right

- **The radial profile.** $\text{score}_F = 0.919 / 0.916 / 0.875 / 0.797 / 0.717$ — better than the previous-generation model at every epoch, and improving with redshift. (These are from the corrected fit; the pre-correction fit gave $0.919 / 0.916 / 0.875 / 0.797 / 0.716$, i.e. the shape half of the problem was entirely unaffected by the benchmark error — see Section 7.2.)
- **Stellar mass growth.** The model's own $M^*(<100 \text{ kpc})$ growth from $z = 2$ to $z = 0.4$ is right to **+0.030 dex** (corrected fit; +0.032 before). The previous generation's kernel needed +0.398 dex — a 13× improvement in a quantity that was previously supplied by a normalisation rather than predicted at all.
- **It generalises.** Held out on 1197 galaxies never used in the fit, the robust scatter of the amplitude residual is identical to in-sample at every epoch (0.1156/0.1202/0.1218/0.1305/0.1507 against 0.1136/0.1178/0.1226/0.1310/0.1530). Seven global parameters do not overfit 1199 galaxies, as one would hope.

5.3 What it gets wrong

Against a purely statistical halo-only regression, it loses.

score_A (1.0 = ties the regression)	z=0.4	z=0.7	z=1.0	z=1.5	z=2.0
on the full 2397-galaxy sample	1.131	1.100	1.059	1.058	1.050
on a complete sample of massive haloes	1.116	1.356	1.339	1.360	1.371

On the sample as selected, the model is 5–13% worse than a regression. On a sample where the selection has been controlled (Section 6), it is **~36% worse at $z \geq 0.7$** . The regression barely notices the restriction — its scatter falls only 3–15% — while the model degrades sharply.

The plain reading: the model is weakest exactly where the data are best, on massive, well-resolved, completely-sampled haloes. That is the deployment case, and it is currently the strongest argument against the model as it stands.

Where in the profile that failure sits. Median $100 \times (\text{model} - \text{truth}) / \text{truth}$ by tercile of $\log_{10} M_h(z=0.4)$, for the adopted model:

tercile	epoch	2.8 kpc	10.2 kpc	27.5 kpc	148.2 kpc
13.00–13.18	$z = 0.4$	–4.0	–2.0	–1.6	–4.1
13.18–13.46	$z = 0.4$	–6.5	–2.2	+0.3	–3.9
13.46–15.14	$z = 0.4$	–12.6	+1.3	+6.3	–3.5
13.00–13.18	$z = 2.0$	–12.6	–9.3	–7.0	–5.1
13.18–13.46	$z = 2.0$	–16.6	–10.0	–7.9	–6.6
13.46–15.14	$z = 2.0$	–24.8	–6.2	–2.8	–4.9

The failure in the most massive tercile is **not** uniform across radius. At 148 kpc the top tercile is no worse than the bottom (–3.5 against –4.1 at $z = 0.4$). What grows is the **misplacement**: –12.6% in the centre together

with **+6.3% at 27.5 kpc**, against -4.0% and -1.6% in the lightest tercile. All three short-listed models show it, so it is a property of the framework, not of one profile family.

Binned by halo mass at the epoch itself, inside the mh-complete sample, the statement sharpens into the single worst thing the model does:

$\log_{10} M_{200c}(z=2)$	n	2.8	4.9	10.2	27.5	79.8	148.2 kpc
12.80–12.91	280	–21.6	–16.6	–11.9	–11.5	–11.6	–12.4
12.91–13.09	280	–27.5	–19.0	–13.6	–12.3	–12.3	–13.0
13.09–14.12	280	–29.2	–16.1	–1.7	+1.5	–2.5	–4.9

In the most massive $z = 2$ haloes the model is 29% too light inside 3 kpc and essentially correct from 10 kpc outward. That is a pure central failure, not a global one — and the two lighter bins fail differently, being uniformly $\sim 12\%$ too light at every radius. The central deficit in the top mass bin grows monotonically with redshift: **-12.6% at $z = 0.4$, -20.4% at $z = 1.0$, -29.2% at $z = 2.0$.**

(The galaxy-to-galaxy scatter is wide — 16th–84th percentile width $\sim 60\%$ at 2.8 kpc — but with $n = 280$ per bin the median is determined to about $\pm 2\%$.)

Plainly: the model cannot build the dense centres of the most massive galaxies at $z = 2$. Those are exactly the compact, early-forming systems the project set out to study, so this is not a peripheral failure — it is the model failing hardest on its target population. It is also the strongest motivation for the compact second channel of Section 9.

5.5 The same failure seen as a density profile: the outskirts are too shallow

The curve of growth is cumulative, so it is dominated by the inner regions and hides what the outer profile is doing. Differencing it into a **surface density** $\Sigma(R) = dM/dA$ and plotting $\log_{10} \Sigma$ against $R^{(1/4)}$ — the coordinate in which a de Vaucouleurs ($n = 4$) profile is a straight line — exposes a second, independent statement of the same defect.

Outer logarithmic slope $d \log_{10} \Sigma / d \log_{10} R$, fitted over 50–148 kpc on the median profile (more negative = steeper):

epoch	truth	model	model – truth
$z = 0.4$	–2.78	–2.74	+0.04
$z = 0.7$	–2.85	–2.78	+0.07
$z = 1.0$	–2.91	–2.81	+0.10
$z = 1.5$	–3.01	–2.86	+0.15
$z = 2.0$	–3.38	–2.90	+0.48

The simulation's galaxies get steeper with redshift; the model's barely change. By $z = 2$ the model's outskirts are 0.48 dex per dex too shallow. In the cumulative view this is nearly invisible — the $z = 2$ curve of growth is only -0.024 dex low at 148 kpc — because a shallow outer slope adds mass in the outermost annuli while the total stays low.

The two high-redshift failures are one statement. Too little mass in the centre *and* too shallow an outer slope both say: **the model's high-redshift galaxies are too diffuse.** It is not that the model makes too little stellar mass at $z = 2$; it makes roughly the right amount and spreads it out too far. That is a much sharper target than “an inner deficit plus an amplitude deficit”, and it points squarely at the deposit size law $R_{50} = f_0 R_{200c}(t_j) (1+z)^b$: the fitted $b = -0.735$ shrinks deposits toward high redshift, but evidently nowhere near enough.

(The truth's own slope is worth noting independently: these galaxies really are close to de Vaucouleurs at $z = 0.4$ — the $z = 0.4$ truth tracks a straight line in $R^{(1/4)}$ across the whole 2–148 kpc range — and they steepen substantially by $z = 2$.)

5.4 The standing structural defect: the model under-fills the centre

Per galaxy, $\log_{10} [M_{*_model}(<R)] - \log_{10} [M_{*_truth}(<R)]$, median over galaxies. Negative means the model puts **less** stellar mass inside R than the simulation does. Measured on the corrected fit:

All 2397 galaxies:

R [kpc]	2.8	4.9	10.2	27.5	79.8	148.2
z = 0.4	-0.034	-0.040	-0.004	+0.008	-0.008	-0.016
z = 0.7	-0.044	-0.039	+0.003	+0.020	+0.009	+0.003
z = 1.0	-0.049	-0.033	+0.007	+0.021	+0.012	+0.007
z = 1.5	-0.073	-0.048	-0.012	+0.002	+0.001	-0.002
z = 2.0	-0.084	-0.065	-0.038	-0.026	-0.024	-0.024

Mh-complete sample (the completeness-controlled galaxies of Section 6):

R [kpc]	2.8	4.9	10.2	27.5	79.8	148.2
z = 0.4	-0.034	-0.040	-0.004	+0.008	-0.008	-0.016
z = 1.0	-0.060	-0.029	+0.016	+0.031	+0.011	+0.003
z = 2.0	-0.133	-0.081	-0.039	-0.030	-0.040	-0.043

Three things this says, none of which is the single number this defect used to be quoted as:

1. **The deficit is inner and it is real**, but at low redshift it is modest: -0.034 dex at 2.8 kpc at $z = 0.4$, i.e. the model is $\sim 7.5\%$ too light there. (An earlier generation of this model was -0.39 to -0.47 dex; the previous internal summary of the present model quoted -0.10 dex, which the corrected fit does not reproduce — the measured value is -0.034 .)
2. **It grows strongly with redshift**, to -0.084 dex on the full sample and **-0.133 dex (26% too light) on the mh-complete sample at $z = 2$** . The model most badly under-fills the centres of high-redshift galaxies — which is exactly the population known independently to be unusually compact.
3. **At low redshift it is genuinely misplaced mass; at high redshift it is also missing mass**. At $z = 0.4$ the outer profile is within 0.016 dex, so the total is roughly right and the error is one of distribution. At $z = 2$ on the mh-complete sample the whole curve is low (-0.043 dex at 148 kpc) with the centre lower still, so there are two errors superposed: not enough mass, and too little of it in the middle.

No profile family fixes the inner part — `gompertz_log`, `moffat` and `sersic` give the same curve — and it is not a resolution artifact in the direction that would explain it away: gravitational softening makes the simulation's cores *less* concentrated than reality, so the true deficit against nature is at least this large.

6. The progenitor-selection confound, and why it mattered

6.1 The problem, stated plainly

Every galaxy in this project was chosen at $z = 0.4$. The high-redshift samples are therefore **not high-redshift samples** — they are the main progenitors of a low-redshift selection. At $z = 2$, a halo of a given mass is in our sample only if it will grow fast enough afterwards to clear the $z = 0.4$ threshold. At low halo mass we keep a small, growth-selected minority; at high halo mass we keep essentially everything.

6.2 Measuring it

Completeness, defined: $N(\text{our sample}) / N(\text{all TNG300 central haloes})$ in a bin of $\log_{10} M_{200c}$ at one epoch. 1.0 means every halo of that mass is in our sample. Computed against the *full* simulation catalogue — 346,000 to 460,000 central haloes per epoch.

The $z = 0.4$ ceiling is **0.781**, set by data-quality flags that are applied to the $z = 0.4$ galaxy and therefore cost the same at every epoch. The **mh-complete cut** is the halo mass at which completeness first reaches 60% of that ceiling and stays there:

	$z=0.4$	$z=0.7$	$z=1.0$	$z=1.5$	$z=2.0$
cut, $\log_{10} M_{200c}$	>13.00	>13.00	>13.00	>12.90	>12.80
galaxies kept	2397	1781	1436	1145	840

At $z = 0.4$ the cut keeps everything, which is the right sanity check: there is no progenitor bias at the selection epoch.

6.3 The result: one of the two “defects” was the survey

The model’s halo-mass tilt at $z = 2$, with parameters frozen:

tilt at $z=2$ [dex/dex]	5 kpc	10 kpc	75 kpc	148 kpc
full sample	−0.135	−0.075	−0.083	−0.094
mh-complete sample	−0.011	+0.066	+0.099	+0.089

It does not shrink — **it flips sign**, and lands on the same positive tilt the mh-complete sample shows at $z = 0.7$ – 1.5 . Removing the incomplete regime makes $z = 2$ consistent with the other epochs instead of an outlier.

Meanwhile the *amplitude* deficit at $z = 2$ gets **worse**, from −0.0255 to −0.0421 dex. So of the two things we thought were wrong with the model at high redshift, one was the survey and the other was under-reported.

6.4 Why we believe it is the selection and not a range effect

A slope measured over a narrower mass range can differ from one measured over a wider range for reasons unrelated to selection. Two further tests close that gap.

The mechanism test. A selection can only bias a residual if the residual depends on the variable the selection truncated. That variable is **future growth**, $G_k = \log_{10} M_h(z=0.4) - \log_{10} M_h(z_k)$ — how much the halo grows *after* the epoch being scored, which is what the $z = 0.4$ selection cuts on and what the model, reading only epoch-local quantities, cannot see. Measured: partial correlation +0.19/+0.22/+0.23/+0.21 and slope **dy/dG = +0.18/+0.16/+0.15/+0.15 dex per dex** at $z = 0.7/1.0/1.5/2.0$. Significantly non-null at every epoch, in the direction required.

The magnitude test. Combining the measured completeness curve with the measured dy/dG under a truncated-Gaussian model predicts the size of the tilt shift, using no fitted model quantity:

tilt shift at 148 kpc	observed	predicted (lower bound)	predicted (truncation-corrected)
$z=0.7$	−0.0148	−0.0170	−0.0443
$z=1.0$	−0.0331	−0.0296	−0.0844
$z=1.5$	−0.0667	−0.0429	−0.1364
$z=2.0$	−0.1830	−0.0542	−0.1918

The observation lies inside the bracket at every epoch. The uncorrected estimate matches where truncation is mildest, the corrected one matches $z = 2$ where truncation is most severe — which is how each should behave if the mechanism is the right one. **The size of the tilt shift needs no physics.**

6.4b A trap in reading fair-sample figures

The mh-complete sample is a different set of galaxies at each epoch. The cut is applied to the halo mass *at that epoch*, so the surviving galaxies are 2397 / 1781 / 1436 / 1145 / 840, and they are progressively more massive:

epoch	n	median $\log_{10} M_h(z=0.4)$ of that set
$z = 0.4$	2397	13.293
$z = 0.7$	1781	13.415
$z = 1.0$	1436	13.489
$z = 1.5$	1145	13.541
$z = 2.0$	840	13.623

So a figure that plots each epoch's mh-complete sample side by side is **not an evolutionary sequence** — most of what changes between the curves is which galaxies are in them. Quantified at 2.8 kpc: the apparent rise in central stellar mass from $z = 0.4$ to $z = 2$ is **+0.247 dex as plotted**, but only **+0.062 dex** when the same galaxies are followed. **Three quarters of it is the changing sample.**

Following one fixed set (the 840 fair at $z = 2$) at every epoch gives the real picture, median $\log_{10} M_*(<R)$:

R [kpc]	2.8	10.2	27.5	148.2
$z = 0.4$	10.744	11.154	11.364	11.540
$z = 1.0$	10.766	11.129	11.289	11.430
$z = 2.0$	10.806	11.044	11.134	11.197

This is inside-out growth, and it is the central physical fact the model has to reproduce. Between $z = 2$ and $z = 0.4$ the outskirts grows by 0.34 dex while the central 2.8 kpc does not grow at all — it is very slightly *higher* at $z = 2$. Galaxies of this mass finished their centres before $z = 2$ and spent the next 9 Gyr adding mass at large radii.

(The small central *decrease* toward low redshift is not a measurement error: TNG's enclosed mass genuinely decreases in 50–58% of these galaxies at ~ 5 kpc, which is why the model carries a measured monotonicity floor rather than a survival term.)

6.5 What is a limitation, not a bug

The framework describes the descendants of massive low-redshift galaxies. The haloes it discards are ones whose growth stalls, whose centrals are typically not massive ellipticals. Two consequences are accepted scope limits:

1. **The model cannot yet be applied to a halo sample selected at high redshift and evolved forward.** It has only ever been calibrated on backward-traced progenitors.
2. High-redshift performance must be *quoted* on the mh-complete sample, not the full one, because the full one flatters (Section 5.3).

7. Two defects in the scoring, and what they changed

Both were found during Stage 3.4 and both changed conclusions.

7.1 A unit bug in the concentration excess

The halo-structure catalogue stores M_{200c} as $\log_{10}(M/M_{\text{sun}})$. Two modules converted it as if it were a *linear* mass in units of $1e10 M_{\text{sun}}/h$, so the reference concentration was evaluated at a mass **2.0006 dex too low**.

Effect on $d\log c$: a spurious 0.15 dex drift with redshift, and a halo-mass slope that changes sign at $z = 0.4$ (-0.053 as coded against $+0.047$ correct).

Scope: 33 of 45 factorial cells read the variable — via E3, via S2a, and via S3. After re-running all 45, `gompertz_log-E2-S2` was still rank 1, but **the four largest upward rank moves in the factorial were all E1+E3 cells** (29 to 13, 34 to 18, 37 to 23, 36 to 22). The earlier verdict “concentration never reaches the top 15” was partly an artifact of feeding it a corrupted variable. Correcting it does not make concentration competitive with the cross-term, but it had not been given a fair hearing.

7.2 One corrupt galaxy inflating the benchmark

Row 181’s measured $M^*(< 100 \text{ kpc})$ collapses from $10^{11.712}$ at $z = 0.4$ to a flat $\sim 10^{7.96}$ at all four higher redshifts — 2.5 dex below the sample’s 0.1st percentile. A main progenitor cannot be 3.76 dex lighter than its own descendant; this is a broken cross-match, not a galaxy.

It sits in the 2397-galaxy set that defines `sigma_A`, and in **none** of the model-fitting subsamples (300, 240, 1199 — all take every second galaxy, and 181 is odd). **So every score_A divided a model error measured on clean galaxies by a benchmark charged for a corrupt one.**

best halo-only regression	z=0.4	z=0.7	z=1.0	z=1.5	z=2.0
as published	0.1047	0.1359	0.1417	0.1456	0.1744
with row 181 rejected	0.1044	0.1120	0.1204	0.1279	0.1611

gompertz_log-E2-S2 score_A	z=0.4	z=0.7	z=1.0	z=1.5	z=2.0
as published	1.128	0.907	0.899	0.929	0.970
corrected	1.131	1.100	1.059	1.058	1.050

This overturned the previous headline. “The model beats the best halo-only regression at four of five epochs” became “it is 5–13% worse at every epoch.”

The damage is bounded to the amplitude side. Row 181 moves the population *shape* loss by **+0.043%**, because the amplitude residual is $\log_{10}(\text{model}/\text{truth})$ — unbounded, and its was 3.76 dex — while the shape residual is computed on a profile normalised to 1, so it lives near unity and no single galaxy can move a 2397-galaxy mean. Every `score_F` stands.

8. What we have learned

8.1 About the physics

1. **A three-line deposition law reproduces the radial profile well.** Seven global parameters, no per-object freedom, `score_F` from 0.92 down to 0.72 with increasing redshift. The profile half of the problem is largely solved.
2. **The mass×redshift cross-term is the single most valuable term.** Without it (E1) the model has a halo-mass tilt of $+0.024$ to $+0.041$ dex/dex; with it (E2) the tilt collapses to $+0.003$ to $+0.012$. Physically: *how* efficiency depends on halo mass must itself change with redshift. This replaced what an earlier generation of the model achieved with an explicit transport term.

3. **Concentration is not competitive with that cross-term**, even after the unit bug was fixed. It improves, but reaches only rank 13 of 45.
4. **No peak in the efficiency's redshift dependence is identified.** E4, the free spline built precisely to find one, is the worst family in the factorial with $\text{ident} \sim 1.5\text{e-}07$ — its parameters are essentially undetermined — for a loss gain of ~ 0.007 .
5. **The profile family barely matters.** Spread across families is 0.05 against 0.08 across efficiency forms. What a deposit is shaped like matters less than how much mass it carries and where it is placed.
6. **The deposition-only assumption is not yet the binding constraint.** A floor computed from the simulation's own non-monotonic behaviour is 0.0456; the models sit at 0.113–0.145 — $2.5\times$ to $3.2\times$ headroom. **The functional forms are the limitation, not the deposition-only premise.**

8.2 About what the model still cannot do

7. **The inner deficit is unsolved and unexplained.** 21% too little stellar mass inside 3 kpc, $\sim 5\%$ too much at 20–70 kpc, total nearly right. Three generations of experiment have failed to remove it by varying the profile family, the objective, or the transport term. What has *not* been tried is a **second, compact deposit channel** — the profile family is the shape of one component, and a second component is a different move.
8. **The model degrades on complete, massive samples** — $\sim 36\%$ worse than a regression at $z \geq 0.7$, against 5–10% on the selected sample. This is an amplitude failure at high halo mass, and it is distinct from the inner deficit. It is currently unexplained and deserves its own investigation.

8.3 About two terms that turned out to be fitting the survey

9. **The size-law conditioning (S3) absorbs selection structure, not physics.** Two independently-constructed mh-complete samples both drive all three of its slopes to near zero from large full-sample values:

parameter	full sample	mh-complete sample A	mh-complete sample B
s_M (mass)	0.0992	0.0032	0.0202
s_c (concentration)	−0.3380	−0.0165	−0.0242
s_f (formation time)	−0.3342	+0.0043	+0.0051

10. **And so, apparently, does E5.** Its whole advantage over E2 was that it closes the $z = 2$ halo-mass tilt — and that tilt is the survey. **E5's four extra parameters buy a fit to a selection artifact.** The case for the seven-parameter E2–S2 is *stronger* after the selection control than before it, which was not the expected outcome.

8.4 About method — the lessons that generalise

11. **A benchmark and the thing it benchmarks must be scored on the same objects.** One corrupt galaxy in the denominator and not the numerator overturned an experiment's headline. When a metric is a ratio computed by two different scripts, assert the object sets match — it is one line.
12. **An unexpectedly good result deserves the same audit as a bad one.** A benchmark degraded by outliers flatters whatever it is compared against.
13. **Rank by a multi-criterion judge, never by the loss.** Demonstrated twice: the loss-optimal cells are precisely the degenerate ones.
14. **Check the completeness of your sample before naming a residual.** A delivery-delay kernel fitted to the $z = 2$ tilt would have absorbed a survey effect and acquired a physical name it had not earned.
15. **Classify which code a change invalidates from what the code reads, not from a list written by hand.** A hand-written list missed S2a; a control cell asserted not to move caught it within three cells.

9. Where the model stands, honestly

What is solid. A seven-parameter, no-pin, no-per-object-freedom deposition law that predicts the radial profile well, gets stellar-mass growth right to 0.03 dex, generalises without a gap, and whose one important term (E2) survives every control we have applied.

What is not. It does not beat a purely statistical halo-only regression at any epoch, and on a completeness-controlled sample of massive haloes it is about a third worse. Two of the extensions that appeared to help — E5 and S3 — appear to be fitting the sample selection rather than the galaxies.

What comes next, in order:

1. Drop S3; treat E5 as suspect for the same reason.
 2. Build the **compact second channel** against the inner deficit, tied to f_{form} so it can fail rather than fit anything. Score it on the fair sample, since the full one is now known to flatter.
 3. Investigate the high-mass amplitude failure — why the model degrades where the data are best. That, not the inner deficit, is what stands between this model and deployment.
 4. A delivery-delay kernel remains a last resort. If built, it must carry an *empirical* halo-mass dependence: $\tau = \eta * t_{\text{dyn}}$ cannot work, because for a $200\times$ critical-density halo $V_{200c} = 10 H(z) R_{200c}$ exactly, so $t_{\text{dyn}} = R_{200c}/V_{200c} = 1/(10 H(z))$ is a function of redshift alone — the halo mass cancels identically, and a mass-independent clock cannot produce a mass-dependent effect.
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Appendix: where things live

item	file
halo record (contains no stellar data by construction) the model, families, truncation	experiments/exp54_unpinned_amplitude/halo.py model.py, experi- ments/exp53_deposition_only/families.py fit.py, hongshao/objective.py
objective, scores, identifiability halo-only regression benchmark the 45-cell factorial and its re-run five-epoch fits	scoreboard.py stage32.py, stage32_rejudge.py stage33.py, stage33_refit.py, stage33_perepoch.py
the selection control QA figures	selection.py qa_figures.py